

Technology Transfer Across Regulatory Regimes: Relational Contracts and State Capacity in Vietnam

Abstract

Technology transfer requires coordination between continuing contractual relationships and obligations distributed across regulatory regimes. Using Vietnam as a theory-informing emerging-economy case, this article connects relational contract theory with administrative capacity to examine that problem. Non-negative matrix factorisation and interpretive coding of thirty-four Vietnamese media, policy and legal-scholarly documents organise seven themes into four dimensions, which generate questions for doctrinal analysis rather than representative estimates of transactional failure. Four findings follow. First, the dominant dimension concerns continuing cooperation and performance, while technology transfer law relies heavily on contractual content requirements and referrals to general law. Second, digital transfers require coordination among transfer, intellectual property, data and AI rules; recognising an algorithm as transferable does not settle the obligations associated with its use. Third, semiconductor capability requires industrial support and negotiated access alongside treaty-constrained compulsory licensing. Fourth, agricultural diffusion depends on extension institutions and valuation capacity beyond bilateral contracting. Functional comparisons distinguish legitimate regulatory specialisation from unresolved contractual defaults and coordination questions. The article proposes a dual-function account of the transfer contract: an arrangement for performance between firms and a potential compliance protocol linking information, responsibilities and regulatory interfaces. This does not make public duties privately negotiable. A pilot inter-agency coordination mechanism and testable implications translate the account into a research agenda. The contribution is analytically transferable to comparable institutional settings, not a claim that Vietnam represents all emerging economies or that the corpus establishes theoretical saturation.

Keywords: technology transfer; relational contracts; regulatory coordination; administrative capacity; computational qualitative analysis; Vietnam.

1 Introduction

Modern technology transfer presents a governance problem that a licence document alone cannot resolve. A transaction may combine continuing technical assistance with rights in an invention, access to data and obligations governing downstream use. The data-access questions examined by Drexler (2017), the complementary assets emphasised by Teece (1986), and the contractual collaboration analysed by Gilson, Sabel and Scott (2009) expose different parts of this problem. The EU data and AI instruments and the US and EU semiconductor frameworks considered below illustrate distinct regulatory responses, not a single international template for transfer contracts. The shared analytical question is how participants can sustain a working transfer relationship across those legal and institutional boundaries.

For an economy seeking both to acquire advanced technology and to develop domestic capabilities, this creates a dual policy task: connect transactions to applicable legal commitments while building the capacity to implement them. International obligations under TRIPS must be distinguished from foreign regulatory or funding requirements that become relevant only within their particular scope. The EU AI Act and US CHIPS legislation are therefore comparators and potentially transaction-relevant regimes, not rules that Vietnam is generally obliged to transplant. Administrative capacity is part of the explanation, not an assumed deficiency of every emerging economy.

Vietnam makes the coordination problem analytically visible through the coexistence of a dedicated technology transfer statute, general private law and recently revised sectoral frameworks. It is used as a theory-informing case, not as a statistically representative member of the Global South. The intended generalisation concerns a mechanism: where transfer requires both continuing cooperation and decisions across distinct regulatory mandates, the usability of the connections between those mandates may matter alongside the content of each rule. Whether that mechanism operates similarly elsewhere depends on legal scope, institutional powers and implementation resources, not merely on sharing an emerging-economy label.

To examine that mechanism, the article begins with published accounts of transfer problems rather than with a list of presumed statutory omissions. A corpus of thirty-four Vietnamese documents was analysed using topic modelling and an interpretive consolidation informed by axial coding. Seven themes were organised into four dimensions, which guide questions put to the statute book. This supplements indispensable doctrinal analysis by making agenda-setting more explicit; it does not suggest that doctrinal research cannot engage with practice or justify its selection of questions.

The move that matters here is one of interpretation. A dimension produced by topic modelling is, on its face, merely a cluster of documents that share vocabulary. Treated that way it tells us about media attention and nothing more. But the documents in this corpus are not neutral reports. They are arguments — practitioners describing disputes they have handled, officials explaining what their agencies cannot do, scholars identifying doctrines Vietnamese law has not adopted. Read as arguments, each dimension carries an implicit claim about what the law should be doing. The analytical task is to recover that claim and put it to the statute.

The relational character of these questions is not a new discovery about contracts. Macaulay (1963) and Macneil (1980) explain why exchange cannot be understood solely through formal promises, while Gilson, Sabel and Scott (2009) examine contractual arrangements supporting collaborative innovation. The contribution here is to connect that perspective to the Vietnamese discourse and to identify where technology-specific rules, general contract law and sectoral regulation must work together. Comparisons with EU data and licensing instruments and US and EU semiconductor legislation serve as functional benchmarks, not models assumed to be superior or directly transplantable.

The resulting argument concerns coordination and the specificity of legal protection, rather than an empty statute book. In particular, a September 2026 analysis must account for Vietnam's Law on Artificial Intelligence No. 134/2025/QH15, the semiconductor chapter of the Law on Digital Technology Industry No. 71/2025/QH15, and Decree No. 101/2026/NĐ-CP replacing the 2018 technology transfer decree. Their existence changes the question from whether the State has responded to whether the different responses provide usable guidance for particular transfer relationships.

Three research questions make that inquiry explicit:

- RQ1.** What claims about the law emerge from the principal themes in the selected Vietnamese technology transfer discourse?
- RQ2.** For each claim, which rules supply an answer, subject to what conditions, and where is a technology-specific default or a connection between regimes not established by this review?
- RQ3.** What does this pattern imply for contractual design and state management, and what further evidence would distinguish legal-design problems from implementation and capability constraints?

The article proceeds as follows. Section 2 connects Vietnamese scholarship with relational contracting, innovation and computational legal research. Section 3 explains the method and its limits. Sections 4 to 7 present the four dimensions, each supported by corpus evidence and examined against the statutory framework and selected comparators. Section 8 distinguishes technology-specific rules, distributed or conditional protection, and questions for which no tailored default is identified. Section 9 concludes.

2 Prior scholarship and the contribution of this study

The study draws on three connected bodies of work: relational contract theory, the governance of innovation, and Vietnamese doctrinal scholarship. Its contribution concerns their application and the transparency of agenda-setting, rather than the discovery of an unstudied subject.

Macaulay (1963) foregrounds informal practices in business exchange; Macneil (1980) situates contractual obligations within continuing social relations. Williamson (1979) offers a distinct, complementary explanation based on transaction-specific investment, uncertainty and the governance of exchange. These approaches should not be collapsed into a single theory. Together they sensitise this analysis to cooperation, adaptation and safeguards against opportunism. Gilson, Sabel and Scott (2009) provide the more direct innovation comparator: formal arrangements can support iterative collaboration rather than merely specify a finished product. A detailed document and a well-governed relationship are therefore not opposites; the question is which provisions make continuing cooperation workable.

The innovation literature also prevents an exclusively contractual explanation. Teece (1986) links commercial returns to appropriability and complementary assets, while Bradley, Hayter and Link (2013) criticise linear accounts of university technology transfer that neglect informal mechanisms and organisational conditions. This suggests a rival explanation for disappointing commercialisation: clearer ownership or licensing rules may be necessary without being sufficient when skills, finance, facilities or organisational incentives are missing. The present corpus cannot determine the relative causal importance of those factors.

The administrative-capacity perspective connects those private relationships to public organisation. The volume edited by Lodge and Wegrich (2014) distinguishes analytical, coordination, regulatory and delivery capacities. Applied here, these categories direct attention to identifying a transfer's legal implications, connecting competent bodies, exercising lawful oversight and supporting implementation. They do not imply that multiple statutes constitute dysfunctional fragmentation. The proposed connection to relational con-

tracting is that performance may depend on both cooperation between firms and workable interactions with public decision-makers.

Figure 1 summarises the proposed connection: one transfer relationship can require both private performance arrangements and coordination with public authorities, without merging the two legal functions.

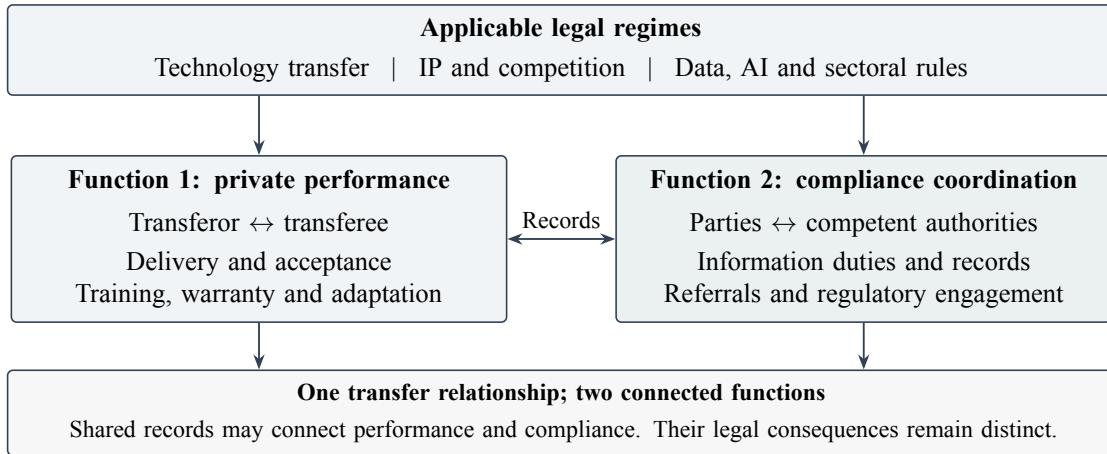


Figure 1: The proposed dual-function account of a technology transfer contract. The links are conceptual, not estimated causal effects. A contract neither binds regulators nor reallocates statutory powers.

The oldest strand is the reform-proposal literature that accompanied the replacement of the 2006 Law. Writing while the amendment was under preparation, Phạm Chí Trung (2016) identified the provisions of the 2006 Law requiring revision, among them the treatment of the contract’s mandatory contents and the relationship between the technology transfer statute and the general law of contract. The 2017 Law adopted some of those proposals and declined others; as Section 8 records, two features of the 2006 Law that this literature had treated as useful — the express designation of the applicable law among the contract’s contents, and a dedicated remedies provision — were removed rather than improved. Hồ Minh Khánh (2024) provides the most direct recent assessment of the resulting regime, examining the contractual provisions of the 2017 Law and concluding that they remain incomplete on several of the same points. The present article agrees with that conclusion and differs in how it is reached: where the existing assessment identifies deficiencies by reading the statute, this study identifies them by reading what participants say they need and then asking the statute whether it supplies it.

A second Vietnamese strand addresses intellectual property and competition. Bùi Thị Hằng Nga (2019) argues against treating every refusal to license by a dominant holder as unlawful, favouring assessment of competitive harm and innovation incentives. The source appears here both as scholarship and as a corpus item; its inclusion in both capacities is not independent corroboration. Section 4.3 distinguishes refusal to license from restrictions within an agreed licence and compares the Vietnamese provisions with the US licensing guidelines and the EU technology transfer block exemption. Importantly, lack of registration does not exclude trade-secret protection, and lack of dominance does not exclude the Competition Law’s rules on restrictive agreements.

A third strand concerns valuation. Nguyễn Thị Minh Phương and Phí Công Thường (2026) discuss legal

and practical difficulties in valuing intellectual assets. Read with Kamiyama, Sheehan and Martinez (2006), this raises a question about usable evidence and valuation capacity, not simply the existence of a legally mandated price formula. Section 7.2 distinguishes negotiated price, mandatory appraisal and professional valuation standards; absence of the second in a particular transaction does not establish absence of the third.

A fourth strand concerns the university and research-institute channel. Phùng Minh Hải (2026) examines the commercialisation of patents held by research institutes and universities; Phan Quốc Nguyên and Tô Nhật Mai (2024) analyse the legal rules governing patent exploitation and management for innovation in Vietnamese universities; and Phan Quốc Nguyên (2020) compares the Vietnamese and United States frameworks for university technology transfer. This literature is the natural comparator for the second theme within the first dimension, and its persistent finding — that the legal framework has improved while commercialisation has not — is echoed by the official assessment quoted in Section 6.1. The 2025 reforms have since altered the terms of that debate, since Articles 24 to 28 of the Law on Science, Technology and Innovation No. 93/2025/QH15 now allocate ownership of results produced with state budget funding, a question this literature had treated as unresolved.

The methodological contribution is a bounded application of computationally assisted agenda-setting. Frankenreiter and Livermore (2020) situate text analysis within a wider range of computational legal methods, which do not all support the same inferences. Following Nelson’s (2020) computational grounded theory framework, this study combines pattern detection with close reading. It then adds doctrinal evaluation: asking whether the law responds to the interpreted problem. That last step is not Nelson’s independent confirmation of an empirical pattern and should not be presented as equivalent to it. Non-negative Matrix Factorisation (Lee and Seung, 1999), comparison with Latent Dirichlet Allocation (Blei, Ng and Jordan, 2003), and the coding framework of Strauss and Corbin (1998) make aspects of selection explicit, but do not remove researcher judgment from corpus construction or interpretation.

3 Method

3.1 Why a qualitative approach, and what it can establish

Computational text analysis can make the selection and organisation of legal material more explicit (Frankenreiter and Livermore, 2020). Here it supports, rather than replaces, doctrinal judgment: topic groupings guide close reading, from which the researcher formulates legal questions. Published discourse supplies accounts of perceived needs, not a direct observation of all parties’ needs or proof that legislation caused the reported difficulties.

The defence of the thirty-four-document corpus rests on this bounded purpose, not on a claim that a small sample becomes statistically representative when analysed computationally. NMF provides a low-dimensional representation for qualitative sorting (Lee and Seung, 1999); LDA is a distinct generative probabilistic model used here as a comparator (Blei, Ng and Jordan, 2003). Neither replaces close reading. The evidence for a legal argument is the attributed passage and the applicable rule, not the algorithmic label alone. This use remains sensitive to sample composition and preprocessing; dimensionality reduction does not remove those limitations.

Nor is *theoretical saturation* a substitute label for representativeness. A defensible saturation claim would require a documented process of iterative sampling and category development showing what additional material changed, and why further sampling ceased to add relevant conceptual properties or relationships (Strauss and Corbin, 1998). Such a record is not available here, and saturation is not claimed. Official provenance or scholarly status establishes neither exhaustive coverage of public debate nor the absence of excluded perspectives. The narrower justification is that these documents provide identifiable accounts from which to formulate and scrutinise legal questions; the sufficiency of that material for a broader theory remains open.

Two limits follow directly and should be stated before the findings rather than after them. The corpus is small, so the proportions reported below describe this corpus and not Vietnamese technology discourse at large; they indicate relative emphasis, not population frequencies. And the corpus consists of published discussion, not of contracts or judgments, so it establishes what participants say is problematic, not what in fact goes wrong in practice. Where the article reports what the law provides, that comes from reading the statutes. Where it reports what practitioners find difficult, that comes from the corpus, and is attributed accordingly.

Figure 2 distinguishes the reported computational procedure, interpretive judgment and legal assessment. A topic label is not a legal finding, and doctrinal assessment does not independently establish a reported problem’s prevalence, especially where the same source supplies both vocabulary and argument.

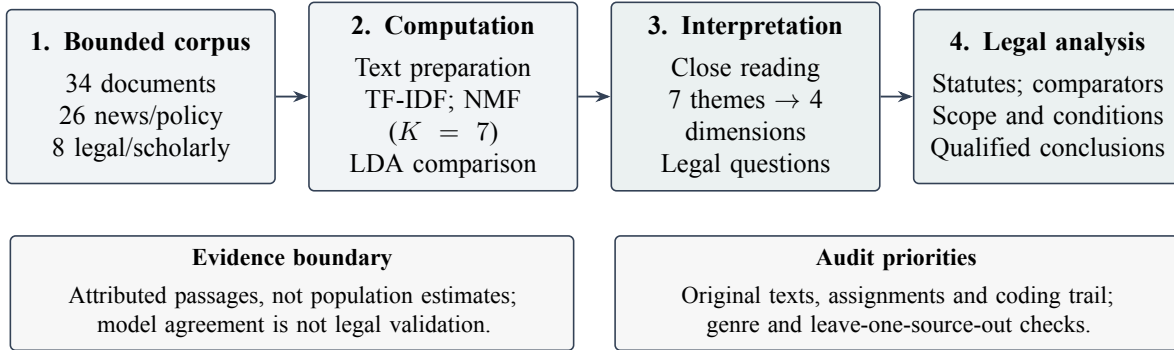


Figure 2: From a bounded corpus to questions for legal analysis. This schematic summarises the reported workflow; it does not represent a new model run or completed robustness tests. NMF and LDA remain distinct models.

3.2 Corpus and coding

The corpus comprises thirty-four Vietnamese-language documents on technology transfer, science, technology and innovation: twenty-six items of news and policy reporting from government portals and the general and specialist press, and eight scholarly or professional legal publications. Each file is treated as one document. Because most items are reproduced without a masthead date, the latest calendar year mentioned, excluding forward-looking policy targets, was used as a provisional dating proxy. The resulting approximate 2012–2026 span is not a verified chronology of publication and cannot support an inference about trends over time. Genre is another source of uncertainty: legal writing and journalism may separate because of register and vocabulary rather than because they express distinct underlying needs.

The available analysis specification reports lower-casing, URL removal, expansion of selected Vietnamese abbreviations, lexicon-based maximum-matching tokenisation with domain multi-word terms, and removal of general and structural stop words. It specifies a TF-IDF representation with minimum document frequency 2 and maximum document frequency 0.9, followed by Non-negative Matrix Factorisation (NMF). These details describe the specified pipeline; they do not substitute for the original scripts, lexicon and matrices needed to audit its execution.

Seven topics were fixed for interpretation, rather than established as a unique statistical optimum. The reported Adjusted Rand Index between the NMF and Latent Dirichlet Allocation (LDA) partitions is 0.448 at seven topics, compared with 0.060 at an automatically selected three. Additional checks using TF-IDF/LSA and BM25 representations are reported in the analysis description, but a complete comparison across topic counts is not available here. The agreement statistics are sensitivity indications, not proof of an algorithm-independent structure or of the stability of every small cluster. With thirty-four documents, topic selection and preprocessing choices can materially affect the partition.

Reproducibility remains a limitation of the present reporting. A complete replication package would need the document inventory and retrieval criteria, reliable publication dates or explicit date uncertainty, Vietnamese tokenisation and stop-word decisions, model settings and random seeds, document–topic assignments, and the coding trail connecting passages to themes and dimensions. The reported agreement statistics do not substitute for those materials, nor do they establish inter-coder reliability or theoretical saturation. The theoretical and comparative sources added for interpretation are not additional corpus documents and do not change the reported counts or percentages.

Each topic was then read qualitatively rather than accepted as a list of words. The analysis uses an axial-coding worksheet informed by Strauss and Corbin (1998) to organise themes and excerpts, but does not complete the paradigm-model assignments of causal conditions, strategies and consequences. Accordingly, “axial coding” here denotes an interpretive consolidation, not a completed grounded-theory explanation. The analytical pair used in the findings is deliberately narrower: what the discourse describes, and what the law provides. Table 1 reports the NMF-based themes; their labels are interpretive, whereas the computational output consists of vocabulary weights and document assignments.

Themes were then consolidated into four higher-order dimensions. The reported average-linkage clustering of NMF topic vectors, using cosine distance and four aggregates, groups Themes 1, 2 and 6; groups Themes 4 and 7; and leaves Themes 3 and 5 as singletons. The interpretive grouping reaches the same partition for a different reason: Themes 1, 2 and 6 raise questions about continuing relationships between identifiable parties, whereas Themes 4 and 7 raise questions about digital technology and state control. Agreement between these two operations is a consistency check, not proof that this is the only defensible grouping. Table 2 records the partition retained for legal analysis.

The percentages state the relative concentration of this corpus. They are not estimates of the frequency of these concerns in Vietnamese technology discourse at large, and nothing in the analysis below turns on their precise values.

Theme 6 warrants a separate robustness caution. The reported document-level comparison assigns the

Table 1: The seven second-order themes

Theme	Subject matter	Documents	Characteristic vocabulary
Theme 1	Technology transfer activity in general	13	transfer of technology, activity, Vietnam, service, organisation, investment, research, foreign
Theme 2	Innovation and higher education	4	innovation, research, application, education, university, symposium
Theme 3	Technology in agriculture	4	agriculture, farmer, rural, production, crop, emissions, artificial intelligence
Theme 4	Artificial intelligence in industry, energy and data	5	data, energy, sector, safety, model, system, operation
Theme 5	The semiconductor industry	3	semiconductor, chip, sector, forum, industry, global, design, engineer
Theme 6	Transfer contracts, intellectual property and competition	3	contract, transfer, law, right, essential, dispute, ownership, obligation
Theme 7	Quantum technology and cybersecurity	2	quantum, list, strategic, network, attack, platform, national

Table 2: The four analytical dimensions

Dimension	Constituent themes	Documents	Share
1. Foundational transfer activity and its legal frame	Themes 1, 2, 6	20 of 34	58.8%
2. Digital technology, data and security	Themes 4, 7	7 of 34	20.6%
3. Semiconductors and industrial capability	Theme 5	3 of 34	8.8%
4. Agricultural technology	Theme 3	4 of 34	11.8%

contract-law assessment by Hồ Minh Khánh (2024) and the practitioner note to one LDA topic, and Bùi Thị Hằng Nga’s (2019) essential-facilities article to another. LDA therefore does not reproduce the three-document NMF cluster as a unit. This diagnostic, like the overall agreement statistic, remains subject to audit against the original assignments. It nevertheless prevents treating overall agreement as confirmation of the theme’s doctrinal label. Theme 6 identifies material for close reading; it does not independently discover or validate an essential-facilities gap. A leave-one-source-out analysis and comparisons within genres would test whether the grouping survives removal of its most doctrinally distinctive source. Those tests are proposed, not reported as completed.

3.3 Statutory material and comparative scope

The legal review uses 16 September 2026 as its cut-off and distinguishes adoption from entry into force. For technology transfer, the starting point is Law No. 07/2017/QH14 as amended through Law No. 115/2025/QH15, read with Consolidated Text No. 32/VBHN-VPQH and Decree No. 101/2026/NĐ-CP. Article 69 of that decree replaced Decree No. 76/2018/NĐ-CP from 1 April 2026, subject to transitional provisions. Article 17 issues the technology lists and Article 18 governs price and payment. The relevant pages of the signed decree were inspected as images; absence of a searchable text layer is not a reason to treat a repealed decree as current law.

The review also draws on the Intellectual Property Law as amended through Law No. 131/2025/QH15; the Competition Law No. 23/2018/QH14; the Civil Code No. 91/2015/QH13; the Commercial Law No. 36/2005/QH11; and the Law on Science, Technology and Innovation No. 93/2025/QH15. The sectoral comparison includes the Data Law No. 60/2024/QH15, the Law on Artificial Intelligence No. 134/2025/QH15, the semiconductor provisions of the Law on Digital Technology Industry No. 71/2025/QH15, and the Law on High Technology No. 133/2025/QH15, effective from 1 July 2026. Valuation must also be read against the Law on Prices No. 16/2023/QH15, including its 2025 amendment No. 140/2025/QH15, and Circular No. 37/2024/TT-BTC on intangible-asset valuation. Consolidated texts are aids to reading, not substitutes for checking subsequent amendments.

Law No. 20/2026/QH16, adopted on 24 August 2026, is recorded as an enacted but prospective change at this cut-off. Its Article 4 amends Articles 41(2) and 54(2) of the Law on Technology Transfer; under Article 5(2), those changes take effect on 1 October 2026, rather than the law’s general commencement date of 1 March 2027. They are not treated here as provisions already in force.

A negative finding is limited to the instruments and particular function examined. Keyword searches can locate provisions, but cannot establish that an obligation is absent: equivalent duties may use different terminology, arise under general law or appear in implementing instruments. Where the review does not establish complete sectoral coverage, the result is an open coordination question, not a finding that the conduct is unregulated. In particular, the paper does not undertake an exhaustive audit of every data-security, cybersecurity, export-control or valuation instrument.

The international comparison is functional and selective. Relational contracting informs questions about continuing performance; US licensing guidance and the EU exemption framework illuminate competitive restraints; EU data and AI provisions distinguish access from downstream compliance; and TRIPS and the

US and EU chips instruments distinguish compulsory access from industrial capability. These sources are not coded into the Vietnamese corpus, do not provide a matched cross-country evaluation, and do not establish the effectiveness of a foreign rule or justify transplantation without regard to institutional capacity. Historical or transitional instruments are identified explicitly, including Regulation (EU) No. 316/2014, replaced by Regulation (EU) 2026/877 from 1 May 2026.

4 Dimension one: a relational problem met by a documentary rule

The first dimension accounts for 58.8 per cent of the corpus — twenty of thirty-four documents — and is the largest grouping, although its size does not determine the legal importance or robustness of every constituent theme. It brings together general transfer activity (thirteen documents), innovation and university commercialisation (four), and the legal literature on transfer contracts (three). Their interpretive connection is a concern with sustaining transfer relationships rather than merely completing a transaction. As Section 3.2 explains, the small legal cluster is less stable across models than its inclusion in this aggregate might suggest.

4.1 What the discourse describes

The clearest statement comes from a practitioner writing on contract disputes, who observes that transfer takes place through a wide variety of instruments — “independent technology transfer contracts, franchise contracts, contracts assigning industrial property rights, contracts for the sale of machinery and equipment accompanied by technology transfer” — and that this diversity has been accompanied by “an increase in disputes between the parties over technology transfer contracts” (Document 33).

The same author then identifies where those disputes come from, and the observation goes to the heart of this article’s argument:

“The rights and obligations of the parties to a technology transfer contract depend largely on what the parties agree in the contract. The law, specifically the 2017 Law on Technology Transfer, sets out only basic provisions concerning the rights and obligations between the parties to the transfer.” (Document 33)

This account locates much of the work of defining the relationship in the parties’ agreement. It supports asking whether they have usable terms and defaults for continuing performance. It does not, without examination of the contracts and disputes themselves, establish that the statutory allocation caused the difficulties described elsewhere in the corpus.

Two of those difficulties are worth quoting because they illustrate what “relational” means concretely. The first concerns payment when the market moves:

“Because the assets and technology are transferred at the moment the market needs them, when the market changes the transferee invokes the ground that the technology has become ‘obsolete’ in order not to perform the payment obligation.” (Document 33)

The second concerns whether the technology was ever fully handed over:

“The transferor must provide training and trial operation, ensuring that products manufactured with that technology are stable, and has an obligation to monitor and supervise for a certain period. After that period, the products manufactured do not meet the requirements, do not satisfy the technological criteria. In such a case, has the transferor inadvertently concealed some know-how? The technology not being transferred completely causes the product...” (Document 33)

Both are relational failures. Neither is a failure to execute a document. In the first, the contract was signed and the technology delivered, and the dispute is about whether an intervening change in market conditions excuses payment. In the second, the documents were exchanged and the training given, and the dispute is about whether what was actually transferred was the whole of the technology or a version with the critical know-how withheld. These are questions about performance over time, and they are the questions the corpus reports as generating litigation.

Administrative fragmentation appears in the same dimension and has the same relational character. A commentary on transfer within foreign direct investment projects notes that three separate groups of state bodies touch technology transfer — those handling investment, those responsible for science and technology, and those responsible for commerce — and that the Law on Technology Transfer contains no provision coordinating them (Document 7). The consequence, in that author’s phrase, is that the science and technology authorities “manage only the treetops”, because by the time a project’s technology content reaches them the investment decision has been taken elsewhere. A second commentary states the problem in more general terms: “legal documents are still duplicative, overlapping, with many loopholes” (Document 31).

Finally, the smallest of the three themes — three documents from the legal literature — supplies a related question about access. One study of the essential facilities doctrine describes an undertaking holding an irreplaceable asset as tending to prevent competitors from entering by withholding access (Document 29). This concerns refusal to deal, which must be distinguished from restrictive terms within a licence already granted. The article supplies a doctrinal question for examination, not a computationally confirmed account of the concerns of the other thirty-three documents.

4.2 What the law provides

The relational interpretation supplies a more precise test than asking whether the statute contains a contract checklist. Macneil (1980) directs attention to continuing cooperation and adaptation, while Williamson (1979) highlights vulnerability after transaction-specific investments have been made. Applied here, the questions are whether parties can verify delivery, adjust performance and protect complementary investments. These are interpretive questions prompted by theory; the corpus does not test either theory causally.

Article 23 of the Law on Technology Transfer specifies fourteen contractual subjects, including the technology, mode of transfer, rights and obligations, price, schedule, warranty, contractual penalties and liability for breach. Article 6 supplies modes of transfer, and Article 27 supplies payment options, now elaborated by Article 18 of Decree No. 101/2026/ND-CP. The latter retains detailed bases for percentage payments, including net selling price, net revenue and profit. Other subjects, notably warranty, receive less tailored

substantive treatment within the contract chapter. A requirement to address a subject should not, however, be equated with a rule that an empty heading satisfies the law.

The template's unfinished warranty and liability clauses illustrate the work left to the drafter, not the legal validity or prevalence of such drafting. Article 22(3) refers to other applicable laws. The Civil Code and Commercial Law do differ, but those differences are not necessarily an irresolvable conflict: Article 4 of each instrument must be considered when identifying the general and specialised rules applicable to a commercial transaction. For example, Article 301 of the Commercial Law generally caps a contractual penalty at eight per cent of the value of the breached obligation, whereas Civil Code Article 418 defers to other relevant laws when they provide otherwise. The Vietnamese concept *phạt vi phạm* should therefore be described as a contractual penalty, not assumed to be equivalent to common-law liquidated damages.

A further distinction concerns the first reported dispute. A claim for an agreed but unpaid price is not the same as a claim for compensation for loss of technological value. The former raises performance and overdue-payment questions; the latter requires attention to causation, proof and mitigation under the applicable liability regime. An allegation of obsolescence does not establish that loss is legally indeterminable or that the transferor has no remedy. Nor can the intellectual property infringement measures in Article 205 simply be imported into contract law. The case-based scholarship represented by Đỗ Văn Đại (2022) identifies the appropriate next evidential setting: analysis of judgments and contractual terms, rather than deduction of judicial outcomes from a template's blank spaces.

The withheld-know-how account nevertheless raises an important design question. Article 23 does not expressly enumerate a technology-specific acceptance procedure linking performance tests, payment milestones and warranty commencement. This is narrower than saying that Vietnamese law has no concept of acceptance. Gilson, Sabel and Scott (2009) show why formal arrangements can support continuing collaboration. For the transactions described here, a suitable model clause could specify demonstrable operating capability, training deliverables, testing responsibilities, corrective work and an independent expert procedure, while allowing adaptation to the technology and protecting confidential know-how. Whether a statutory default is preferable to professional model terms remains a policy question.

4.3 Restrictive terms: overlapping protection and its limits

The doctrinal theme raises two distinct questions: which restrictions may be included in a licence, and when access may be compelled despite refusal to license.

Clause 13 of the standard form offers alternatives concerning territorial exclusivity and onward transfer. Their legal assessment depends on the rights licensed, the actual restraint and the applicable contract and competition rules. A restriction is not unlawful merely because it limits the licensee's freedom of action.

Article 144(2) of the Intellectual Property Law prohibits specified unreasonable restrictions in industrial property licences, including certain constraints on improvements, export, sourcing and validity challenges; Article 144(3) makes the prohibited terms void. Its scope cannot be reduced to registered rights. Trade secrets are industrial property under Article 4(4), with rights established without registration under Article 6(3)(c), subject to protection conditions including Article 84. The relevant boundary is whether the licensed

subject matter qualifies for the applicable protection, not simply whether it has been registered.

Competition law supplies more than a dominance-based route. Articles 24 and 27 govern abuse of dominance; separately, Articles 11 and 12 address restrictive agreements. Article 12(4), in particular, covers specified vertical agreements where they have or are likely to have a significant competition-restricting effect. Thus a non-dominant transferor licensing unpatented technology is not necessarily outside competition law.

The residual question is more modest: where neither an applicable IP prohibition nor the competition-law test resolves the clause, does general contract law supply sufficiently predictable guidance about its proportionality? This is a potential gap in tailored guidance, not proof that no law applies. A refusal-to-license doctrine also does not by itself determine the validity of every grant-back, sourcing or territorial clause in a negotiated agreement.

The international benchmarks clarify the choices. The US Department of Justice and Federal Trade Commission's 2017 licensing guidelines treat licensing as often procompetitive and do not presume that an IP right confers market power; their coverage includes trade secrets and know-how. The EU's Regulation (EU) 2026/877, applicable from 1 May 2026, provides conditional block exemption for qualifying technology transfer agreements, including qualifying secret, substantial and identified know-how. Its market-share conditions, hardcore restrictions and excluded restrictions structure assessment; an agreement outside the exemption is not automatically unlawful. The accompanying 2026 Technology Transfer Guidelines support individual analysis. Regulation (EU) No. 316/2014 is a historical comparator, not the current September 2026 regime, subject to the successor's transitional provisions.

Article 40 of TRIPS supplies the broader comparative starting point: Members may address licensing practices that abuse intellectual property rights and adversely affect competition, consistently with the Agreement. The coexistence of transfer, IP and competition rules is consequently not, by itself, evidence of a defective regulatory arrangement. The question is whether a particular restriction can be assessed under a clearly identified route, with intelligible conditions, consequences and institutional responsibilities. Neither TRIPS nor the EU comparison establishes that ordinary Vietnamese transfers necessarily fall outside all protection.

These approaches do not justify making every restrictive clause void or reproducing EU thresholds in Vietnam. They support distinguishing restraints that protect transfer-specific investment from restraints that unnecessarily foreclose competition. Drafting guidance could explain the interaction of Article 144 with Articles 11–14 and 24–27 of the Competition Law, and use worked examples of improvement, grant-back, sourcing and territorial clauses. The proposal is an interpretive inference from the comparison, not an empirical finding that foreign exemptions increase Vietnamese technology diffusion.

For compulsory access, Bùi Thị Hằng Nga's (2019) caution about innovation incentives remains pertinent. Articles 145 and 146 of the Intellectual Property Law provide a separate patent-specific route, with Article 147 allocating decision-making authority. The statutory grounds include unsuccessful reasonable licensing negotiations and prohibited restrictive conduct, but not every ground depends on a competition-law infringement. The existence of different competent bodies warrants examination of referral and decision procedures; it does not by itself prove that coordination is legally absent or has failed in practice.

5 Dimension two: digital transfers across regulatory regimes

The second dimension accounts for 20.6 per cent of the corpus and combines two themes: industrial applications of artificial intelligence, energy and data (five documents), and strategic technology including quantum computing and cybersecurity (two).

5.1 What the discourse describes

What is striking about this dimension is how consistently it frames digital technology in terms of risk and control rather than transaction. A representative document observes that artificial intelligence “poses new challenges regarding information security, data protection, privacy, disinformation and national sovereignty”, and that the requirement is “not only to apply AI to defence but to master the technology proactively, to build autonomous capability and to prepare in advance for the risks” (Document 14). The same document notes that AI “can also be exploited to automate attacks, generate falsified content, develop malware”, and that the rapid development of quantum computing “threatens public-key algorithms, requiring a stepwise transition to post-quantum cryptography in order to protect state secrets, strategic data, cryptographic sovereignty and national security”.

The vocabulary here — sovereignty, secrets, attack, mastery — belongs to security policy, not to contract law. But the underlying subject is transfer: who may obtain a model, a dataset, or a cryptographic capability, and on what conditions.

5.2 What the law provides

Four legal functions must be separated: recognising an object of transfer, determining rights in it, regulating access or cross-border movement, and allocating duties when an AI system is supplied or deployed.

Article 4(1)(b) of the Law on Technology Transfer, amended by Law No. 115/2025/QH15, expressly includes models, algorithms, software, information and data. That recognition does not itself establish ownership, make every dataset freely transferable or override another statute’s restrictions.

Law No. 131/2025/QH15 addresses different questions within the Intellectual Property Law. Article 6(5) delegates rules on rights in subject matter created using AI; Article 7(5) permits specified uses of lawfully published, publicly accessible material for research, experimentation and AI training, subject to protection of legitimate interests. These provisions should not be described as a comprehensive data-transfer licence. Lemley and Casey (2021) argue for a US fair-use approach to machine learning, while Samuelson (2023) examines the copyright disputes raised by generative AI. They provide context for the training question, not authority that Vietnamese permission to train a system authorises every subsequent disclosure or transfer of the underlying data.

Nor is the IP statute the only Vietnamese response to AI risk. The Law on Artificial Intelligence No. 134/2025/QH15 took effect on 1 March 2026. It differentiates developers, providers and deployers, classifies systems by risk, and provides transparency and incident-related obligations (Articles 3 and 9–12). Article 7(3) also prohibits data activities that contravene data, personal-data, IP and cybersecurity law. The Data Law No. 60/2024/QH15, effective from 1 July 2025, separately addresses data governance, including

cross-border transfer and processing in Article 23. Their coverage and exceptions must be examined for a concrete transaction; neither is replaced by the technology transfer contract.

The transfer-control lists must also be assessed in their current form. Article 17 of Decree No. 101/2026/NĐ-CP issues encouraged, restricted and prohibited lists in Annexes I–III and provides for periodic or exceptional revision. Whether a particular dual-use model or dataset falls within a list, another sectoral restriction or neither requires classification of the actual object, use and transfer destination. This study does not establish a categorical screening vacuum for digital technology. The concern supported by the discourse is the need to make the relationship among those regimes intelligible to transfer participants.

Protection must likewise be analysed by legal category. A computer program, an original compilation, a qualifying trade secret and raw data do not attract identical rights. Failure to satisfy Article 84’s trade-secret conditions does not necessarily eliminate contractual confidentiality obligations or all other protection. Article 4’s broad definition of transferable objects is therefore not evidence that every such object has an exclusive property right, nor that every object without registration is unprotected.

5.3 Functional comparison: access is not ownership or compliance

Drex1 (2017) distinguishes creating property rights in industrial data from designing access arrangements that support competition. The EU Data Act, Regulation (EU) 2023/2854, illustrates the latter approach: Articles 4 and 5 address user and third-party access to certain connected-product and related-service data, Article 13 addresses specified unfair business-to-business data terms, and Article 43 limits the database right in its defined field. It is not a general right to obtain any dataset or algorithm. Its application is phased under Article 50, and its data-access rules operate alongside personal-data and trade-secret protections.

The EU AI Act, Regulation (EU) 2024/1689, addresses a different function. It entered into force on 1 August 2024, which must be distinguished from the phased application of its obligations. Article 53 requires providers of general-purpose AI models to maintain documentation, supply relevant information to downstream system providers, adopt a copyright-compliance policy and publish a training-content summary, subject to its specific exceptions. These are regulatory duties, not a general transfer of ownership in algorithms or a compulsory licence to all training material. This comparison concerns the specified mechanism, not a claim that every AI Act obligation applies to every system at the same date.

The inference for Vietnam is consequently not that AI and data rules sit in the “wrong” statute. Specialisation across statutes can be appropriate. A more defensible proposal is coordinated guidance identifying the parties’ regulatory roles, lawful data access, permissible uses, onward disclosure, documentation, security responsibilities and incident cooperation. Such guidance should explain which obligations remain mandatory despite contractual allocation. It would address the relational concerns in the corpus without assuming that recognising a new transferable object supplies all of its governance rules.

6 Dimension three: industrial capability and access to technology

The third dimension is the smallest — three documents, 8.8 per cent — and draws attention to industrial capability. Its small size requires particular caution about generalising from the reported concerns.

6.1 What the discourse describes

The semiconductor documents discuss capability, not contracts: the sector as “the technological backbone of every economy”, Vietnam’s position in global design and manufacturing, the supply of engineers, the role of national innovation institutions (Documents 15, 17). The third document in the theme is more critical, reporting official data on an ecosystem with over 3,800 technology startups and 74,000 information technology firms, alongside the assessment of a Ministry of Industry and Trade official that “commercialisation of intellectual assets remains limited” despite an improved legal framework (Document 4).

The policy instruments discussed in the adjacent theme confirm the framing. Decision No. 21/2026/QĐ-TTg of 30 April 2026 promulgated Vietnam’s list of ten groups of strategic technology and the accompanying list of key strategic technology products, covering artificial intelligence, semiconductor chips, green hydrogen, quantum technology and low-orbit satellites, described as a step towards “concentrating resources on core and foundational technologies” (Document 16).

6.2 What the law provides

The sector must be read through both industrial policy and intellectual property law. Decision No. 21/2026/QĐ-TTg identifies strategic technologies and products, while the Law on High Technology No. 133/2025/QH15 provides the high-technology framework. Crucially, Chapter III of the Law on Digital Technology Industry No. 71/2025/QH15 is devoted to semiconductors. It is therefore inaccurate to describe the compulsory-licensing provision as the only binding sector-specific rule.

Article 146(1)(b) of the Intellectual Property Law nevertheless matters for access to patented semiconductor technology. It limits compulsory licensing to public non-commercial use or remedying anti-competitive conduct. The limitation reflects Article 31(c) of the TRIPS Agreement, adopted in 1994; it is not simply a Vietnamese drafting anomaly that an industrial-policy amendment can freely remove. Compulsory licensing also does not automatically supply the tacit know-how, manufacturing expertise or other complementary assets needed to make use of a patent.

6.3 Capability-building beyond compulsory licensing

Teece (1986) explains why access to an invention and the capability to commercialise it are distinct. Grindley and Teece (1997) examine licensing and cross-licensing in semiconductors and electronics, providing a closer comparator for negotiated access than a general call for compulsory licensing. Those studies support attention to complementary capabilities and contractual arrangements; they do not establish which mechanism will succeed in Vietnam.

The US CHIPS and Science Act of 2022 (Public Law 117-167), particularly its semiconductor funding and incentives provisions, and the EU Chips Act, Regulation (EU) 2023/1781, illustrate capability-oriented instruments. The latter supports design capacity, pilot lines and competence centres (Articles 4–5). These are not general mandates requiring patent owners to open their rights. US semiconductor incentives also operate with national-security conditions, including constraints on certain technology licensing and joint research involving foreign entities of concern under the statutory framework. The comparison therefore does not support a simple opposition between foreign openness and Vietnamese restriction.

The governance implication is that strategic technology policy requires proactive capability-building, such as targeted support and shared technical facilities, alongside appropriately bounded licensing and competition remedies. Compulsory access alone cannot supply the complementary resources needed for effective use; conversely, industrial support does not eliminate legitimate competition concerns. This is an inference about the mix of policy instruments, not evidence that a particular subsidy or foreign institutional design will succeed in Vietnam.

The policy inference is to connect Vietnam's existing sectoral incentives with voluntary licences, training, pilot facilities and access to complementary technology, subject to competition and security rules. This could include transparent access terms for publicly supported infrastructure and measurable capability-building commitments in supported projects. The treaty-constrained compulsory-licensing route remains distinct. Whether licensing costs, workforce shortages or access to manufacturing facilities is the dominant constraint requires evidence beyond the three documents in this dimension.

7 Dimension four: a transfer channel that contract law barely touches

The fourth dimension, 11.8 per cent of the corpus, concerns agricultural technology and shows the limits of a contract-centred account of technology transfer. Its share of the corpus does not determine the social importance of the issues involved.

7.1 What the discourse describes

The agricultural documents describe diffusion rather than transaction. Artificial intelligence in agriculture is “still quite new” in Vietnam but attracting government and enterprise interest; the benefits described are higher yields, healthier crops and livestock, disease control, and “organising data for farmers” (Document 21). The actors are ministries, agricultural associations, extension centres and conferences; the instruments are national strategies and programmes. Where the first dimension is populated by contracting parties and their disputes, this one is populated by institutions and their programmes.

7.2 What the law provides

Article 52(5) of the Law on Technology Transfer provides an administrative channel for recognising agricultural technical advances encouraged for transfer. A contract-centred account must therefore accommodate institutions that disseminate knowledge without a conventional bilateral licence. Fuglie, Gautam, Goyal and Maloney (2020) emphasise the interaction of innovation supply with farmers' ability and incentives to adopt technology. The inference here is that extension, skills and access conditions deserve assessment alongside ownership and contract drafting, not that the corpus proves these institutions perform adequately.

Valuation requires a separate distinction. Article 27(1) leaves price to agreement, while Article 27(3) addresses audit or appraisal in specified state-capital and related-party situations. Article 18 of Decree No. 101/2026/ND-CP elaborates price and payment, including appraisal and tax-authority requirements for particular transactions. Neither negotiated price nor the absence of mandatory appraisal in another transaction means that no valuation standards exist. Article 2(18) of the amended transfer law links technology appraisal

to Vietnamese Valuation Standards, and Circular No. 37/2024/TT-BTC specifically addresses intangible assets.

Kamiyama, Sheehan and Martinez (2006) review cost-, market- and income-based approaches and the difficulties of valuing IP in imperfect markets. These alternatives support a different reform priority from imposing a single statutory formula: make the assumptions and evidence behind a valuation transparent. For agricultural transfers, analysts should distinguish the technology's expected benefits from complementary inputs and adoption conditions, and state uncertainty about comparables and projected income. Standardised disclosure of assumptions, access to professional appraisal and appropriately protected transaction evidence are possible responses. Their usefulness, and the extent of any current shortfall, require examination of valuation reports and transactions rather than inference from Article 27 alone.

This valuation question cuts across dimensions rather than constituting a uniquely agricultural omission. University inventions, industrial know-how and agricultural technologies can all pose difficulties of comparability and uncertain future income. What distinguishes the agricultural dimension is the institutional route to adoption and the conditions under which users can benefit, not the absence of a separate statutory price formula for agriculture.

8 The current state of the law

Reading the four dimensions against the statutes produces the map in Table 3. It identifies the legal route for each question, rather than counting every referral or absence of a technology-specific provision as a regulatory failure.

Figure 3 visualises four cross-statute interfaces around the transfer contract, whose dedicated framework remains the Law on Technology Transfer and its implementing decree. It is a selective guide to Table 3, not an exhaustive inventory. The competition interface includes restrictive agreements as well as abuse of dominance; the digital interface distinguishes data governance, AI compliance and the semiconductor framework. The overlapping circles represent questions that may arise in the same transaction, not rules that invariably apply together or evidence of failed coordination.

The categories are: *transfer-framework rule*, where the Law on Technology Transfer or its implementing decree supplies the relevant requirement; *distributed or conditional protection*, where other laws or eligibility conditions determine the answer; and *no tailored default identified*, where the review identifies no technology-specific default for the precise question, without excluding general law. An *open coordination or implementation question* is used where the corpus raises a concern but this review does not establish the adequacy or complete coverage of the relevant rules. None of these classifications establishes how courts or agencies perform in practice.

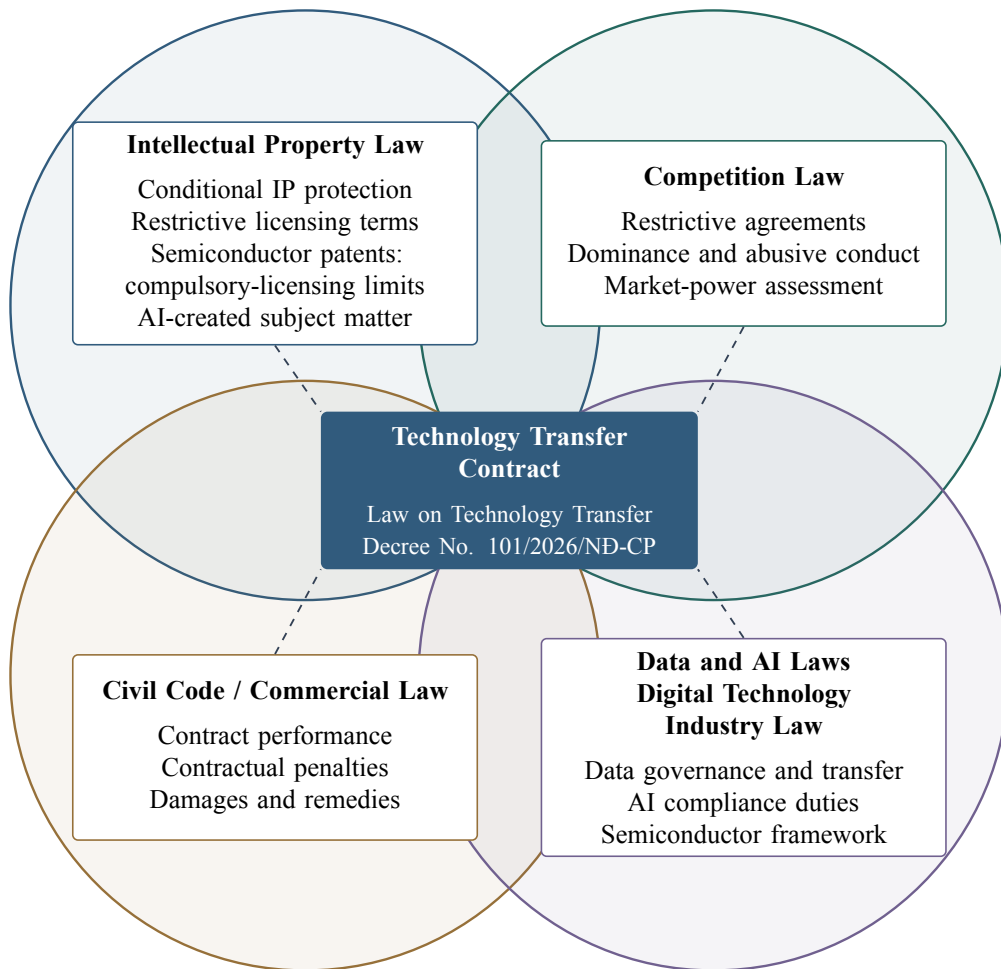


Figure 3: Regulatory distribution around a technology transfer contract: selected interfaces from Table 3. Dashed links indicate connections subject to legal scope and eligibility conditions. The circles are conceptual, not proportional to regulatory burden; their overlap does not establish institutional silos or a particular form of polycentric governance.

Table 3: What the law provides on the questions the discourse raises

Question raised by the discourse	Dimension	Where the answer is found	Classification
What must a technology transfer contract contain?	1	Law on Technology Transfer, Articles 22 and 23	Transfer-framework rule
By what modes may technology be handed over?	1	Law on Technology Transfer, Article 6 (five modes)	Transfer-framework rule
How may the price be paid?	1	Law on Technology Transfer, Article 27(2); Decree No. 101/2026/NĐ-CP, Article 18	Transfer-framework rule
How is agricultural technology diffused?	4	Law on Technology Transfer, Article 52(5)	Transfer-framework rule
What counts as strategic technology?	3	Decision No. 21/2026/QĐ-TTg; Law on High Technology No. 133/2025/QH15	Distributed protection
Which law governs a contract with a foreign element?	1	Civil Code, Articles 664 and 683	Distributed protection; conflict-of-laws analysis
Are restrictive terms permissible in an industrial property licence?	1	Intellectual Property Law, Articles 4(4), 6(3)(c), 84 and 144(2)–(3); qualifying trade secrets need no registration	Conditional protection
May a dominant transferor impose abusive restrictions?	1	Competition Law, Articles 24 and 27	Conditional protection
May a refusal to license a patent be overridden?	1	Intellectual Property Law, Articles 145–147; ground-specific conditions	Conditional protection
What limits compulsory licensing of semiconductor patents?	3	Intellectual Property Law, Article 146(1)(b); TRIPS, Article 31(c)	Treaty-constrained protection
Where is semiconductor capability addressed?	3	Law on Digital Technology Industry No. 71/2025/QH15, Chapter III; high-technology framework	Distributed regulation; implementation not tested
Is an algorithm or a dataset protected?	2	Applicable copyright, trade-secret and contractual rules; protection depends on the subject matter	Conditional protection; no general ownership inference

Continued on next page

Table 3 (continued)

Question raised by the discourse	Dimension	Where the answer is found	Classification
Who owns subject matter created using an artificial intelligence system?	2	Intellectual Property Law, Article 6(5); implementing provisions require separate assessment	Delegated regulation; no complete ownership conclusion here
May lawfully published data be used to train artificial intelligence systems?	2	Intellectual Property Law, Article 7(5); separate data-law obligations	Conditional permission
Who owns the results of research funded from the state budget?	1	Law on Science, Technology and Innovation No. 93/2025/QH15, Articles 24 to 28	Distributed regulation
What remedies follow a breach of the contract?	1	Article 22(3) referral; Civil Code and Commercial Law, including their Article 4 application rules	Distributed protection; classify transaction and remedy
Can restrictions by a non-dominant transferor be controlled?	1	Competition Law, Articles 11–14; IP and contract rules where applicable	Conditional protection; dominance is not always required
How should improvements to know-how be allocated?	1	Agreement and applicable IP rules; distinguish qualifying trade secrets from unprotected information	Open default-design question
How should delivery, acceptance and warranty commencement be linked?	1	Article 23 content requirements and applicable performance rules	No tailored acceptance default identified in the contract chapter
How is uncertain contractual loss proved?	1	Applicable Civil Code or Commercial Law liability rules; distinguish unpaid price from damages	Evidential question; no remedial vacuum established
How are digital transfers and downstream AI responsibilities governed?	2	Decree No. 101/2026/NĐ-CP, Article 17 and annexes; Data Law; Law on Artificial Intelligence; other sectoral rules	Distributed regulation; transaction-specific coordination
What supports valuation of a private transfer?	1, 4	Article 27; Decree No. 101/2026/NĐ-CP, Article 18; Circular No. 37/2024/TT-BTC	Agreement plus valuation framework; capacity not tested

Three observations connect this map to the theoretical and comparative analysis.

Formal documentation and relational governance are complementary. The first dimension makes delivery, adaptation and continuing cooperation visible, but does not demonstrate that general contract remedies fail. Relational theory instead supplies criteria for evaluating particular arrangements: whether technical performance can be tested, responsibility allocated and collaboration maintained. An acceptance protocol or coordinated model terms may improve those functions without requiring a new mandatory rule for every transaction.

Conditional protection is not a legal vacuum. Trade secrets need no registration, restrictive-agreement rules need not depend on dominance, and general contract law supplies remedies outside IP infringement. The EU exemption framework and US licensing guidelines make these distinctions especially important: licensing restrictions can enable investment as well as restrict competition. The reform question is whether the boundaries and legal consequences are predictable, not whether all restrictions should be prohibited.

Coordination and capability are distinct from rule creation. Digital transfers combine ownership, access, security and downstream compliance questions; semiconductor transfers combine negotiated rights with industrial capabilities; agricultural diffusion combines extension and adoption with valuation. The selected comparisons show different ways of assigning these functions, not that all should be moved into the technology transfer statute. A failure of implementation or a shortage of technical capacity would require different evidence and different interventions from a missing contractual default.

The reform record must therefore be read across instruments, not measured by counting amendment annotations in a single chapter. Law No. 115/2025/QH15 and Decree No. 101/2026/NĐ-CP altered the transfer framework, while AI, digital-industry, high-technology and valuation instruments address adjacent functions. Earlier law remains useful for comparison: the 2006 transfer law expressly listed applicable law among contractual subjects and contained a dedicated remedies provision in Article 27. Their removal in 2017 is a reason to examine the clarity of referral to general law, not proof that remedies disappeared. A transaction-level study is needed to establish which remaining difficulties arise from legal design and which from drafting, institutional practice or limited complementary resources.

8.1 Implications for state management

Regulatory distribution becomes problematic fragmentation when relevant decisions, information or responsibilities cannot be connected in a way that permits lawful performance. The presence of several statutes or agencies alone does not establish that condition. Drawing on Lodge and Wegrich (2014), the institutional question concerns the capacity to coordinate distinct mandates while retaining the expertise and accountability attached to each. The timing concern in Document 7 suggests examining when technology information reaches investment, science-and-technology and other competent bodies, not merely which body eventually receives a document.

Figure 4 translates *networked coordination* into a proposed pilot: a single point of contact routes each case to all relevant competent bodies, while each retains its statutory powers. The issue groups are illustrative, not a new allocation of jurisdiction. Named contacts, reasons and service targets would make handovers visible

without displacing statutory time limits, substantive tests or rights of review.

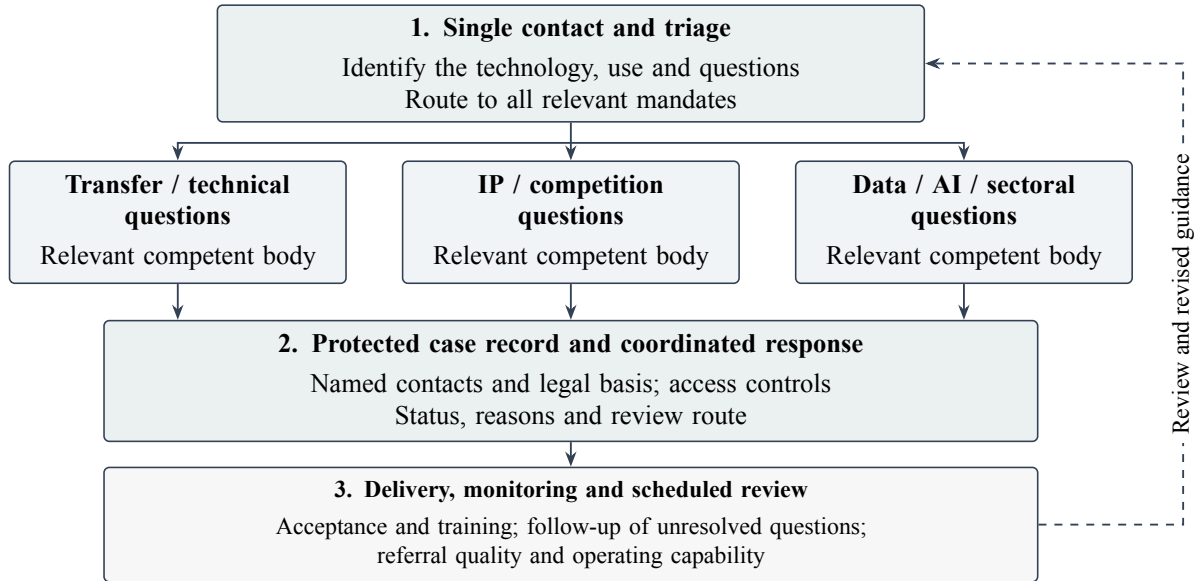


Figure 4: Proposed inter-agency coordination pilot, not an existing legal procedure or a single approval route. Solid arrows show case handling; the dashed loop shows scheduled feedback and revision of guidance, not suspension of binding duties.

The shared record would specify lawful purpose, access permissions, correction responsibilities and limits on retention and reuse; an inter-agency platform would not make confidential know-how or personal data generally accessible. Administrative registration would remain distinct from technical acceptance, commercial viability and adjudication. A time-limited pilot would test whether coordination reduces duplication or creates another bottleneck, whether referrals remain contestable and whether smaller firms can use the process. Staffing, lawful information access and accountability are therefore conditions of the proposal, not benefits assumed to follow from adding a portal.

Evaluation should separate processing activity from transfer outcomes. Candidate measures include repeated information requests, referral time and unresolved jurisdictional questions, alongside documented acceptance, training completion and the recipient's capacity to operate the technology. These are proposed indicators, not observations generated by this corpus. Whether coordination improves outcomes must also be examined against the complementary-resource explanation identified by Teece (1986) and Bradley, Hayter and Link (2013): finance, skills and facilities may remain binding constraints even where legal guidance is clear.

8.2 Testable implications

The relational account yields two propositions for subsequent investigation. First, if uncertainty about delivery is important, otherwise comparable transfers with explicit performance tests, correction procedures and payment milestones should show fewer unresolved delivery disputes than transfers relying on generic descriptions. Second, if coordination is important, projects receiving earlier and clearer inter-agency referral should face fewer repeated compliance requests or late classification changes. Contract files, acceptance

records, administrative correspondence and interviews would permit examination of these propositions. Neither follows causally from the present topic model.

Rival explanations and contrary cases are essential. Well-resourced parties may both draft better contracts and complete transfers more successfully; complex technologies may generate more referrals and more disputes regardless of administrative quality. Cases in which general contract law resolves a dispute despite sparse technology-specific drafting, or in which extensive contractual detail fails to prevent an operational breakdown, would help test the proposed mechanism rather than merely confirm its vocabulary. At the corpus level, withholding a dual-role scholarly source and checking genre-specific groupings would similarly test whether a legal question persists without the source that supplied its doctrinal formulation.

9 Conclusion

This study uses Vietnam to examine a problem of wider relevance: how a continuing technology transfer relationship can operate across distinct private-law and public-regulatory mandates. Its international contribution is not a claim that Vietnam represents other emerging economies. It is an account of connections that can be examined elsewhere: between delivery and adaptation, between contractual rights and mandatory duties, and between regulatory specialisation and administrative capacity.

The theoretical contribution is a bounded extension of the application of relational contract theory. Macaulay (1963), Macneil (1980) and Gilson, Sabel and Scott (2009) direct attention beyond isolated promises to continuing cooperation. The dual-function account in Figure 1 adds a proposed *compliance protocol* to the contract's organisation of private performance: records, information duties and regulatory engagement can connect the two functions without making the State a contractual party or allowing private agreement to alter public powers and duties. Whether these arrangements support one another is the proposition to be tested through the safeguards and pilot in Section 8, not a new legal status established by this study.

The findings support a qualified relational critique. Content requirements are not a substitute for workable arrangements for delivery and continuing cooperation, but neither is referral to other statutes equivalent to non-regulation. Vietnam's AI and data legislation, semiconductor provisions, updated transfer decree and intangible-asset valuation standards rule out several categorical claims of statutory silence. The more defensible concerns are the specificity of contractual defaults, the clarity of interfaces among regimes, and the practical ability to use the rules that exist.

Five priorities follow as proposals for evaluation, not as reforms whose effects the corpus proves. First, develop technology-sensitive model terms for acceptance, training, warranty and adaptation, and test whether a statutory default is needed. Second, clarify restrictive-term assessment across IP and competition law, including trade secrets and agreements between non-dominant firms. Third, provide coordinated guidance on AI and data transfers that distinguishes lawful access, permitted use, security and continuing regulatory duties. Fourth, connect semiconductor incentives with negotiated technology access, skills and shared facilities while respecting TRIPS Article 31(c). Fifth, improve access to valuation expertise and evidence, including for agricultural and university transfers, rather than assume that a single compulsory price formula would solve commercialisation problems.

For state management, these priorities favour an intelligible route through existing responsibilities over another layer of general approval. The proposed coordination protocol and outcome indicators in Section 8 translate the relational argument into questions that an administrative pilot could examine. Its value would lie in helping a transfer reach demonstrable operating capability, not merely in increasing the number of documents processed.

The limitations remain substantial. Thirty-four published documents cannot establish representative needs, causal effects, the prevalence of defective contracts or institutional failure. In particular, Theme 6 is a small grouping potentially affected by genre, which the reported LDA comparison does not reproduce as a unit; it cannot serve as independent computational validation of the essential-facilities argument. Additional references do not cure the incomplete replication materials identified in Section 3.2. The next stage requires a documented corpus and coding trail, followed by contracts, judgments, valuation reports and interviews that can distinguish legal-design problems from implementation and capability constraints. The 2026 reforms provide a useful temporal reference point for that work, not by themselves a causal identification strategy.

The legal map is bounded by the instruments and functions reviewed as at 16 September 2026. In particular, the technology transfer amendments in Law No. 20/2026/QH16 commence on 1 October 2026. Future use of the map requires checking effective dates, implementing instruments and transaction-specific scope rather than relying on the date of a consolidated text alone.

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